Regression Trees

Regression Trees

- The tree will search for all combination of predictors and cutoff value to decide the best split
- In Regression tree, the best split is the split that minimizes

$$\underbrace{\sum_{i:\mathbf{x}_i \in R_1(j,s)} (y_i - \hat{y}_{R_1})^2}_{\text{RSS of obs. in left branch}} + \underbrace{\sum_{i:\mathbf{x}_i \in R_2(j,s)} (y_i - \hat{y}_{R_2})^2}_{\text{RSS of obs. in right branch}}$$

- $\hat{y}_{R_1}$ and $\hat{y}_{R_2}$ are the means of the responses falling in to the left branch and right branch, respectively.

Example

continuous

	X_1	X_2	Y
A	1	0	1.2
B	2	1	2.1
C	3	2	1.5
D	4	1	3.0
E	2	2	2.0
F	1	1	1.6

Using the RSS to decide the best split among

- Split 1: Region 1 $X_1 < 4$, Region 2 $X_1 \geq 4$
- Split 2: Region 1 $X_2 < 2$, Region 2 $X_2 \geq 2$

Split 1

$\boxed{x_1 < 4}$

yes

$y = 1.2$
2.1
1.5
2.0
1.6

no

$y = 3.0 \rightarrow \bar{y}_2 = 3.0$

$\Rightarrow RSS_2 = 0$

$$\bar{y}_1 = \frac{1.2 + 2.1 + 1.5 + 2.0 + 1.6}{5} = 1.68$$

$$RSS_1 = (1.2 - \bar{y}_1)^2 + (2.1 - \bar{y}_1)^2 + (1.5 - \bar{y}_1)^2 + (2.0 - \bar{y}_1)^2 + (1.6 - \bar{y})^2$$

$$= \boxed{.548}$$

$$RSS = RSS_1 + RSS_2$$

$$= \boxed{.548}$$